Deep Dive: Granular Instrumental Variables

Heterogeneous Loadings and Optimal Weighting

Macro Lab Group — Technical Supplement

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1 Introduction

Slides 11 and 12 move from the "Simple Model" (where every firm responds identically to the macroeconomy) to the "Full Model." This is crucial because, in reality, superstar firms often have different sensitivities to interest rates, exchange rates, or global demand than small firms.

2 Slide 11: The Full Factor Model

2.1 The Equation

We start with the demand equation for firm i:

$$y_{it} = \phi p_t + \lambda_i \eta_t + u_{it} \tag{1}$$

where λ_i is the **factor loading**.

2.2 Intuition: What is a Factor Loading?

A factor loading λ_i measures firm i's *sensitivity* to a common macro shock η_t.

- **Example:** If η_t is a surge in energy prices, an airline like Delta might have a high λ_i (very sensitive), while a software firm like Microsoft might have a low λ_i.
- **Heterogeneity:** In the simple model, we assumed $\lambda_i = 1$ for everyone. In the full model, we acknowledge that large firms might be more (or less) sensitive to macro trends than the average firm.

2.3 Derivation: Why the Simple GIV Fails

The simple GIV is defined as $z_t^{simple} = y_{St} - y_{Et}$. Let's derive its components using Equation 1:

1. **Size-weighted average (y_{St}):**

$$\begin{aligned} y_{St} &= \sum S_i y_{it} = \sum S_i(\phi p_t + \lambda_i \eta_t + u_{it}) \\ &= \phi p_t \underbrace{\sum S_i}_{1} + \eta_t \underbrace{\sum S_i \lambda_i}_{\lambda_S} + \underbrace{\sum S_i u_{it}}_{u_{St}} \\ &= \phi p_t + \lambda_S \eta_t + u_{St} \end{aligned}$$

2. **Equal-weighted average (y_{Et}):**

$$\begin{aligned} y_{Et} &= \frac{1}{N} \sum y_{it} = \phi p_t + \eta_t \underbrace{\frac{1}{N} \sum \lambda_i}_{\lambda_E} + u_{Et} \\ &= \phi p_t + \lambda_E \eta_t + u_{Et} \end{aligned}$$

3. **The Resulting GIV:**

$$z_t^{simple} = y_{St} - y_{Et} = (\lambda_{\mathbf{S}} - \lambda_{\mathbf{E}})\eta_{\mathbf{t}} + (u_{St} - u_{Et}) \tag{2}$$

The Problem: If large firms have different macro-sensitivities than small firms ($\lambda_S \neq \lambda_E$), then the instrument z_t^{simple} is correlated with the common shock η_t. It is no longer a "pure" idiosyncratic shock; it is "leaky."

3 Slide 12: The Solution (Optimal Weights)

3.1 The Goal: Orthogonality

We want to choose a new set of weights, Γ_i, to construct a valid GIV: $z_t = \sum \Gamma_i y_{it}$. For this to be a valid instrument for the supply equation, it must be uncorrelated with the macro shock η_t.

3.2 Derivation of the Orthogonality Condition

Substitute Equation 1 into the definition of the weighted GIV:

$$\begin{aligned} z_t &= \sum \Gamma_i(\phi p_t + \lambda_i \eta_t + u_{it}) \\ &= \phi p_t(\sum \Gamma_i) + \eta_t(\sum \Gamma_i \lambda_i) + \sum \Gamma_i u_{it} \end{aligned}$$

To ensure z_t contains *only* idiosyncratic shocks (u_{it}), we must satisfy two conditions:

1. $\sum \Gamma_i = 0$: This removes the endogenous price p_t.
2. $\sum \Gamma_i \lambda_i = 0$: This removes the common macro shock η_t.

3.3 Why not just Firm Size (S_i)?

You asked: "Aren't weights given by firm size?"

- **Power vs. Validity:** Firm size (S_i) gives the instrument its **power**. We want the instrument to be large, and only large firms can create aggregate "granular" ripples.
- **The Conflict:** If we use $z_t = \sum(S_i - 1/N)y_{it}$, we have power, but we lack validity if S_i is correlated with λ_i (e.g., if big firms are more sensitive to macro shocks).
- **The "Tilt":** We start with firm sizes S_i but then "tilt" or project them until they are perpendicular (orthogonal) to the macro-sensitivities λ_i.

3.4 Mathematical Solution (Projection)

In matrix form, we want Γ to be as close as possible to S but satisfy $\Gamma \perp \lambda$. This is a standard projection problem:

$$\Gamma = S - \lambda(\lambda'\lambda)^{-1}\lambda' S \tag{3}$$

Intuition: We take the firm sizes and subtract the portion of size that is explained by macro-sensitivity. What remains is "Pure Granular Weight."

4 Summary

- **Factor Loadings** (λ_i): Firm-specific sensitivities to the macroeconomy.
- **Validity**: We cannot use raw size if size correlates with macro-sensitivity.
- **Optimal Weights** (Γ_i): The "tilted" versions of size weights that satisfy $\sum \Gamma_i \lambda_i = 0$.